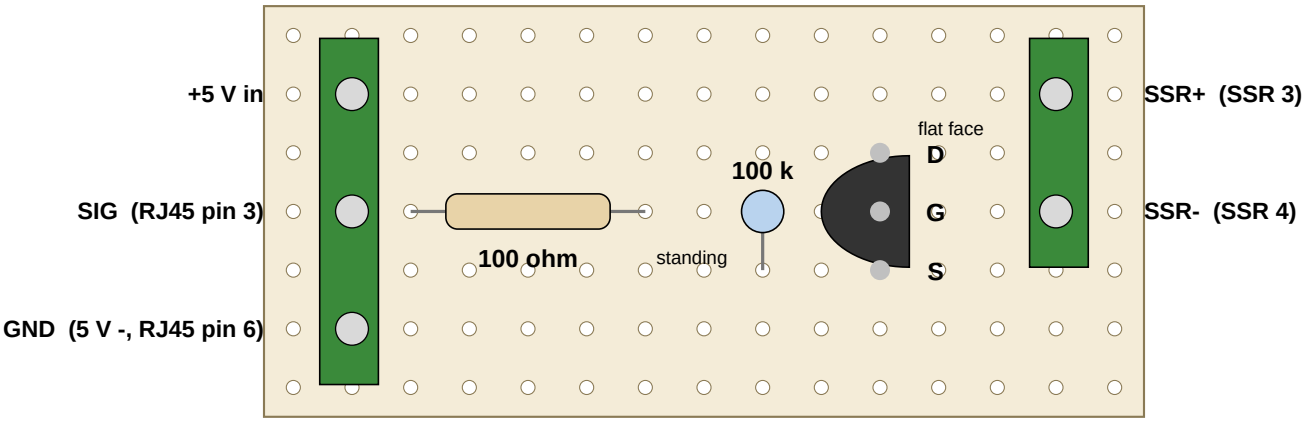
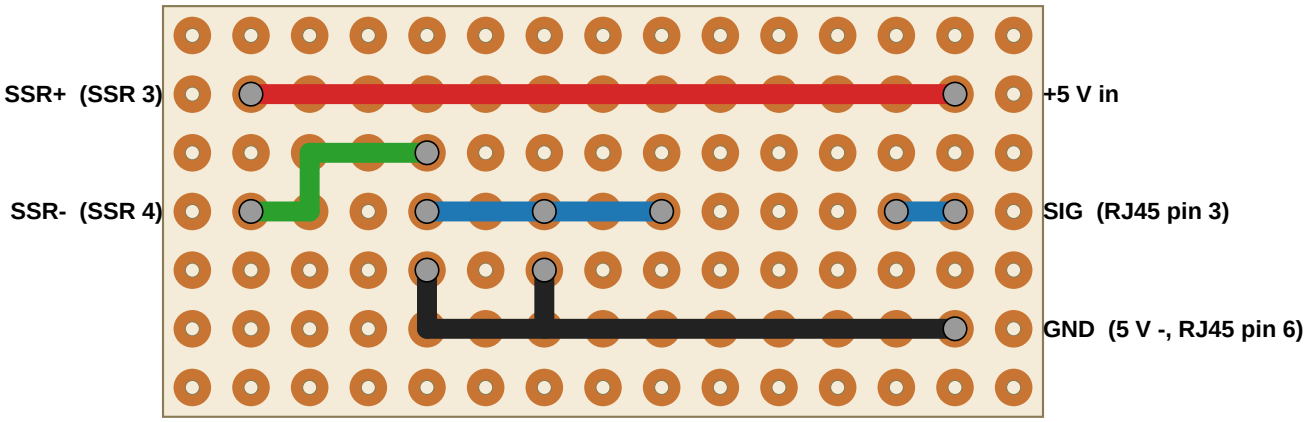
Pellevoisin: pump SSR driver on perfboard

2N7000, 100 ohm, 100 kohm, a 3-way and a 2-way screw terminal (5.08 mm). Board 15 x 7 holes, 0.1 in pitch.

Component side (from above)

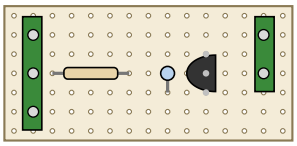


Solder side (board turned over left to right): coloured lines are solder bridges or bare wire



Actual size, component side

this bar is 1 in / 25.4 mm
if not, the printer scaled it



Order of work: terminal blocks first, then the 2N7000 (legs short, flat face towards the output block), then the 100 ohm lying flat, then the 100 k standing: body over the gate-row hole, its lead bent down into the hole below.
Bridges on the solder side: +5 V straight along its row to SSR+; signal to the 100 ohm; the 100 ohm's far end, the 100 k's top and the gate joined; the 100 k's foot, the source and the GND row joined; the drain across to SSR-.
Keep the gate row apart from the drain: the hole between them stays empty.
Check with the meter before mains: SSR+ to GND reads the 5 V; gate to source about 3.1 V with the pump on, 0 off.
This layout is for the 2N7000 (S G D). The RFP30N06LE's legs are G D S: do not drop it into these holes.
Mount on nylon standoffs, on the low-voltage side, near the SSR's input terminals and away from its mains ones.